

Project Design Phase - Problem-Solution Fit

Date	Team ID	Project Name	Maximum Marks
18 June 2026		ShopEZ E-Commerce Platform	2 Marks

1. Problem-Solution Fit Canvas (10 Blocks)

1. Customer Segment [CS]

Urban digital consumers, tech enthusiasts, and bargain shoppers seeking specific appliances, accessories, and groceries.

2. Jobs-to-be-done / Problems [J&P]

- Finding quality items within budget.
- Checking out securely without tedious forms.
- Tracking order fulfillment state from placement to delivery.

3. Triggers [TR]

- Social media advertisements for discounts.
- Need for immediate replacement of a home appliance or grocery item.

4. Emotions: Before/After [EM]

- **Before:** Frustrated by cluttered interfaces, slow loading times, and ambiguous pricing.
- **After:** Relieved by transparent billing, fast responsive filtering, and a clean minimalist dashboard.

5. Available Solutions [AS]

Traditional slow loading e-commerce platforms, multi-vendor sites with high complexity, or physical brick-and-mortar stores.

6. Customer Constraints [CC]

- Limited digital payment options.
- Varying internet bandwidths.
- Trust concerns with online credit card transactions.

7. Behaviour [BE]

Users frequently filter products by price-low/high or rating, add items to the cart, and prefer review confirmations before checking out.

8. Channels of Behaviour [CH]

- **8.1 Online:** Mobile web browsers, social shopping channels, and desktop e-commerce portals.
- **8.2 Offline:** Visiting physical retail outlets for price comparison.

9. Problem Root Cause [RC]

Monolithic architectures that fetch massive payloads without client-side pagination/filtering or proper RESTful API design, causing high latency.

10. Your Solution [SL]

ShopEZ: A React SPA client optimized with client-side state management (`AuthContext`, `CartContext`) connected to a fast Node/Express backend with MongoDB, featuring robust routing, JWT-based security, and dynamic filters.

2. References

- [IdeaHackers Problem-Solution Fit Canvas](#)
- [Medium Guide by Epicantus](#)